Peritoneal Dialysis Effluent Diagnostic Biomarker & Subtyping Framework

Workflow replicating the matrisome-based biomarker discovery pipeline of Sun et al. (2026)

1. Discovery Cohort (GSE125498)

Platform: Illumina HumanHT-12 v4 beadchip
Samples: N = 33 peritoneal effluent cell pellets
Groups: 13 Long-term PD (LPD) vs 20 Short-term PD (SPD)
Background universe: 11,741 unique annotated genes

2. Differential Expression Analysis (limma)

Linear models & empirical Bayes moderated t-statistics
Thresholds: |log2FC| >= 0.585 (1.5-fold) & nominal P < 0.05
Identified: 813 total DEGs (496 Up in LPD, 317 Down in LPD)
Genome-wide Benjamini-Hochberg FDR < 0.05: 143 genes

3. Matrisome Filtration & WGCNA

Naba reference matrisome: 1,027 genes (428 in universe)
Matrisome DEGs: 44 genes (31 Up, 13 Down; 1.71x enrichment, P=0.0024)
WGCNA: 5,001 variable genes clustered into 14 dynamic modules
Trait correlation: Top pro-fibrotic module MEmagenta (r=0.48, P=0.0047)

4. Multi-Algorithm Machine Learning Selection

Candidate feature pool: 38/39 ECM-DEG ∩ non-grey WGCNA genes
• LASSO L1-regularized logistic regression (11 selected)
• SVM-RFE 10-fold recursive feature elimination (10 selected)
• Random Forest mean decrease in Gini importance (10 selected)
Majority-vote consensus (>=2/3 algorithms): 8 core biomarkers

5. Consensus Panel & Clinical Nomogram

Diagnostic panel: TNFSF15, ADAM19, FLT3LG, NAT6, ECM1 (+ LTB, EBI3)
Apparent 5-gene ridge model C-index / AUC: 0.9808 (95% CI: 0.94-1.00)
Validation: Equal-count risk quintiles with exact binomial CIs
Decision curve analysis (DCA) & Clinical impact curve

6. Microenvironment & Immune Infiltration

ssGSEA enrichment across leukocyte and stromal cell markers
Differential infiltration: LPD vs SPD with BH-FDR adjustment
Immune-immune cell correlation architecture
Spearman rank correlation: Diagnostic genes vs immune signatures

7. Functional Networks & Pharmacogenomics

GeneMANIA functional association network (20 interactors)
DSigDB candidate perturbational drug signatures (adjusted P < 0.05)
Single-gene GSEA: KEGG pro-fibrotic pathways per biomarker
Biological role: ECM remodeling, cytokine signaling, cell adhesion

8. Molecular Subtyping & External Audits

Unsupervised consensus clustering on LPD cohort (k=2 / k=3)
Proportion of Ambiguous Clustering (PAC) & PCA separation
Subtype-specific biomarker expression & immune score divergence
External audit: GSE62928 (EPS biopsy), GSE248762 & GSE130888

Fig. 1. Study design and computational workflow.

Integrative pipeline combining transcriptomic profiling, matrisome annotation, co-expression network analysis (WGCNA), three-algorithm machine learning feature selection, clinical risk nomogram modeling, and molecular subtyping.